The Value of a Calibrated Error Bar Is Real Where the Parameter Is Identifiable: An End-Stage Synthesis of Trust, Value of Information, and Action in Quantitative IVIM Imaging

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Abstract

A deployed quantitative-MRI parameter map carries an error bar; a clinician who acts on it faces three questions in order — can the bar be *trusted*, is a trusted bar *worth* anything to the decision, and where does acting on it *break*? This perspective synthesizes a multi-project intravoxel-incoherent-motion (IVIM) uncertainty program into one arc, trust $\rightarrow$ value of information $\rightarrow$ action, and argues that the program answers the three questions as a single chain. It makes no new measurement. The synthesis terminates in a sharp, defensible negative anchored on the pseudo-diffusion coefficient D^*: D^* is *cross-modally orphaned*. It is unidentifiable from its own signal — a Cramér–Rao identifiability bound that grows $\sim 6\times$ across the D^* range and reaches the regime-bin scale in the high-D^* tercile (segmented IVIM, SNR 10–100); it is un-scalable to its own repeatability — on real same-day test–retest data its interval width does *not* track scan–rescan scatter (Spearman $r = -0.17$, $p = 0.13$, $n = 76$; 95% bootstrap CI $[-0.39, +0.05]$, spanning zero), where the well-identified diffusion coefficient D does ($r = +0.60$); and it is only weakly and *cohort-inconsistently* tied to an independent perfusion readout (DCE K^{trans}: $r = 0.389$, $p < 0.001$ in one cohort but null in another). The end-stage caution is therefore not "uncertainty quantification fails" but the sharper claim its own machinery earns: the value of a calibrated error bar is real and quantifiable exactly where the parameter is identifiable, and D^* marks the identifiability wall where trust, value, and action all terminate together.

Provisional status. This is an end-stage synthesis assembled before its in-program anchors have published. Claims marked P depend on an unpublished anchor; the load-bearing spine (the D^* thread) rests on in-repository, reproduced anchors and survives regardless. Submission is held until the trust and value anchors publish (Sec. 7).

Keywords: intravoxel incoherent motion (IVIM); uncertainty quantification; value of information; identifiability; pseudo-diffusion; conformal prediction; decision-making.

1 Introduction: three questions a clinician actually asks

Quantitative MRI increasingly ships not just a parameter map but an *error bar* on every voxel. The premise of uncertainty quantification (UQ) in imaging is that this error bar is actionable — that a clinician, or an automated triage rule, can use it to decide whether to trust a value, escalate to another modality, or spare a patient an intervention. That premise hides three distinct questions, and they must be answered in order:

1. **Trust.** Can the reported interval be believed — does it mean what it says?
2. **Value of information.** *Given* a trusted bar, is it worth anything to the decision the clinician is actually making?
3. **Action.** When the bar is deployed and acted upon, where does it break?

A research program is most honest when its separate results turn out to answer one question each, and chain. This perspective makes that chain explicit for a multi-project IVIM UQ program: a calibration ruler (the *trust* anchor), a decision-theoretic pricing of a trusted bar (the *value* anchor), and two independent accounts of where deployment fails (the *action* anchors). We add no new experiment. We argue a synthesis, and we let the synthesis fall where the evidence pushes it.

Where it pushes is onto a single parameter. IVIM models the diffusion-weighted signal as a biexponential, $S(b)/S_0 = f\, e^{-bD^*} + (1-f)\, e^{-bD}$, separating tissue diffusion D from a fast pseudo-diffusion compartment D^* attributed to microvascular perfusion. D^* is the parameter the program keeps hitting a wall on, in every one of the three questions. We use it as the spine of the argument and call the resulting thread *"D^* cross-modally orphaned"*: the program's whole apparatus works for D (and the perfusion fraction f) and collapses for D^*. Crucially, this is an *identifiability* statement, not a claim that UQ is futile. The distinction is the point of the paper.

2 Trust: the calibration ruler

Before an error bar can have decision value it must be trustworthy: the reported interval must have the coverage it advertises. The trust anchor (Fashion [1]) [P] builds a model-agnostic *calibration ruler* — a check of whether a reported IVIM uncertainty is honest — and shows that a skew-aware posterior (an MCMC quantile interval) restores *marginal* coverage where symmetric $\pm\sigma$ intervals are miscalibrated. In its retooled, boundary-railing-first form the ruler reports this as a scoped secondary finding under an honest Cramér–Rao baseline, with the load-bearing residual being a *conditional* failure concentrated in the high-D^* regime.

The reading that matters for the arc: trust is achievable for D and f, and for D^* only marginally — with a residual high-D^* hole. That hole is the seam the rest of the arc pulls on. (This claim is [P]: Fashion's calibration behaviour must survive review; it is a *contextual* anchor, not part of the load-bearing spine of Sec. 5.)

3 Value of information: pricing a trusted bar

Suppose the bar can be trusted. Is it worth anything? A calibrated interval is not automatically decision-relevant; its value is whatever it changes about the action taken. The value anchor (Minos [2]) prices exactly this for a treat / spare / escalate decision, and — importantly for a paper built before publication — its *theory half* (the Plumbline results) is machine-verified and publication-independent. Three results carry the section.

The decision–calibration gap. The interval scale that is optimal for the *decision* departs from the scale that is optimal for *coverage* whenever the posterior is skewed and the cost is asymmetric. To leading order the gap is $G = \frac{1}{6}\,|z^*(\lambda)|\,\gamma$, with skewness γ and an asymmetry term $z^*(\lambda)$ set by the cost ratio λ (Plumbline Theorem 1). A symmetric problem ($\gamma = 0$) or a symmetric cost ($\lambda = 1$) closes the gap; IVIM is neither.

The value of information is second-order. The gain from re-calibrating the bar to the decision, rather than to coverage, is $V = \mathrm{EU}(\tau^*) - \mathrm{EU}(\tau_{\mathrm{stat}}) = \frac{1}{2}|\mathrm{EU}''(\tau^*)|\,G^2 = O(\gamma^2)$ (Plumbline Proposition 3 — the *Delphi* result). The value is *quadratic* in the gap: a calibrated bar changes the decision's value only above a curvature-set "worth-it" floor. This is a leading-order, small-gap statement and we scope it as such — it characterizes the regime near the decision boundary where the quadratic approximation holds, not arbitrarily large mis-scalings.

The label-free detectability floor. A label-free validity monitor can catch *observable* distribution drift but is provably blind ($\mathrm{AUC} = \frac{1}{2}$) to a *hidden* channel that induces the same regret (Plumbline Theorem 2). This is the formal case that label-free monitoring cannot, alone, certify a deployed bar; labeled spot-checks are required.

The reading for the arc: a trusted, calibrated bar has real, quantifiable decision value — but only where the gap is appreciable, and a label-free monitor cannot certify it by itself. The *theory* here is SOLID (independent of publication); Minos's *applied* half, which consumes the trust and action anchors, is [P]. No applied number from Minos is load-bearing for the spine below.

4 Action: where acting on the bar breaks

When the bar is deployed and acted on, two *independent* walls appear — one about the interval's size, one about which parameter it is even possible to size.

Wrong size (Lethe). On real same-day scan–rescan data (ACRIN-6698 [3], $n \approx 76$), a conformal interval calibrated for accuracy is far too narrow to cover real test–retest variation of D (Lethe [4])[P]. This is not a contradiction: an interval scaled for *measurement accuracy* is *expected* to under-cover *repeatability*, because accuracy coverage and repeatability coverage are different targets and cannot be conflated. Precision is not coverage. The honest-limitation verdict is sharply scoped and survives as a caution about *which* coverage a deployed interval actually provides.

Wrong parameter (Gauge): an identifiability wall. D^* is essentially unidentifiable per voxel (Gauge [5]). The argument is a Cramér–Rao one and we are careful to frame it as such: the Cramér–Rao lower bound (CRLB) is the smallest variance any unbiased estimator can achieve from the IVIM b-value curve at a given operating point — a statement about *local identifiability*, not an impossibility proof. Scoped to the segmented biexponential IVIM scheme and the SNR band 10–100, the CRLB on D^* grows by a factor of ~ 6 across the physiological D^* range (reproduced in this work, Fig. 1; reproduced factor 5.7×), and in the high-D^* tercile the CRLB standard deviation reaches $\sim 1.12\times$ the width of that tercile — the regime is, in this acquisition, unresolvable. That conformal interval widths track the CRLB ($r = 0.77$) confirms the intervals are honest about this: they widen exactly where identifiability fails. What cannot be guaranteed label-free is coverage *conditional* on a latent axis the data do not identify — the IVIM instance of the impossibility of distribution-free conditional coverage [6], and a concrete instantiation of the hidden, undetectable channel of Sec. 3.

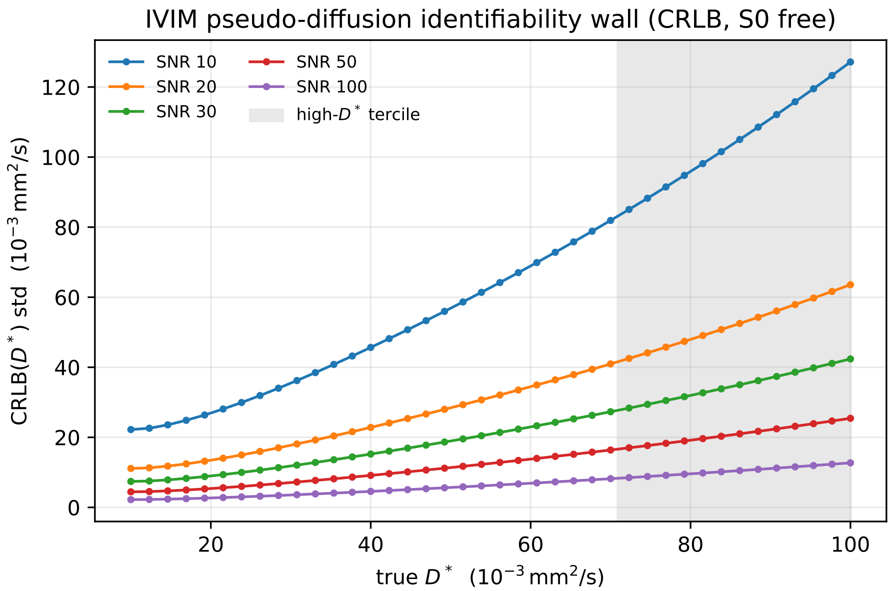

Figure 1: The pseudo-diffusion identifiability wall, reproduced. Cramér–Rao lower bound on D^* (standard deviation, S_0 jointly estimated) versus true D^* across the SNR band, for the segmented biexponential IVIM scheme. The bound grows $\sim 6\times$ across the D^* range and, in the shaded high-D^* tercile, reaches the scale of the tercile width itself: the regime is unresolvable per voxel. This is an identifiability (CRLB) limit scoped to this acquisition and SNR band, not an impossibility proof. Self-contained re-derivation (`scripts/crlb_wall.py`); cross-checked against the committed Gauge anchor.

The reading for the arc: acting on the bar is safe for the parameters the program can identify and size; it is *not* safe for D^*, which sits behind an identifiability wall a label-free monitor cannot see — so labeled repeatability spot-checks are mandatory, not optional.

5 The thread: D^* is cross-modally orphaned

D^* is where trust, value, and action fail *together*, and the failure extends beyond IVIM, across modalities. Three independent observations form the spine. The first two are **in-repository, reproduced anchors** (the load-bearing core); the third is corroborating external literature.

(1) Unidentifiable from its own signal. D^* cannot be recovered per voxel from the IVIM b-value curve: the CRLB grows $\sim 6\times$ in the high-D^* regime and reaches the regime-bin scale there (Sec. 4, Fig. 1). This is reproduced from first principles in this work, not merely cited.

(2) Un-scalable to its own repeatability. On the real same-day test–retest arm, the D^* interval width does *not* track scan–rescan scatter: Spearman $r = -0.17$ ($p = 0.13$, $n = 76$; 95% BCa bootstrap CI $[-0.39, +0.05]$, *spanning zero*) — a null. By contrast the well-identified D tracks its own scatter at $r = +0.60$ (95% CI $[+0.42, +0.72]$, strictly positive). The D^* null is reported *with its interval*, not as a bare point estimate, precisely because a null is only meaningful with its uncertainty; the CI's spanning zero is the load-bearing fact. We stress the caveat that this is a region-level, whole-tumor-ROI repeatability-tracking association on unregistered same-day repeats

— not an in-vivo coverage, accuracy, or calibration claim. The D^* negative is *evidence for* the identifiability thesis, not a gap in it.

(3) Weakly and inconsistently anchored to an independent perfusion measurement. If D^* measured perfusion, it should correlate with an independent perfusion readout. The cross-modal correlation of D^* with DCE-MRI K^{trans} is, at best, weak and cohort-dependent: $r = 0.389$ ($p < 0.001$) in one rectal-cancer cohort [7] but *non-significant* in another [8], which also reports only moderate D^* reproducibility (ICC = 0.55). We frame this honestly as *weak and cohort-inconsistent* — significant in one cohort, null in the other — *not* uniformly "non-significant"; both are cited. Tellingly, the composite $f{\cdot}D^*$ correlates better ($r = 0.533$), consistent with f — not D^* — carrying the recoverable perfusion signal. This strand is corroborating; the load-bearing spine is (1)–(2).

The synthesis. D^* is orphaned: a trusted ruler, a priced decision, and a sized interval all work for D and f and collapse for D^* — unidentifiable from its own signal, un-scalable to its own repeatability, and only weakly and inconsistently tied to an independent perfusion readout.

6 End-stage caution

The defensible end-stage statement is not that uncertainty quantification fails. It is the sharper claim the program's own machinery earns: *the value of a calibrated error bar is real and quantifiable exactly where the parameter is identifiable*, and D^* marks the identifiability wall where trust, value, and action all terminate. Operationally this is a triage rule, and we are careful not to overclaim it at the boundary: trust the well-identified diffusion map D; read high-D^* as unresolved rather than as a measured perfusion value; price the bar's decision value only where the decision–calibration gap is appreciable (Sec. 3); and, because the failing axis is one no label-free monitor can see, treat labeled repeatability spot-checks as mandatory. None of this is a claim that D^* "works" once these cautions are applied — the spine is an honest negative, and we let it stand as one.

7 Provisional discipline and submission hold

This synthesis was assembled *before* its in-program anchors published. We treat that as a discipline, not a disclaimer. Every claim that depends on an unpublished anchor is marked P. Every external number traces to a verified primary source. The load-bearing spine of Sec. 5 — the D^* identifiability wall and the D^* test–retest null with its bootstrap CI — rests on in-repository, reproduced anchors and is *self-contained*: the trust (Fashion) and value (Minos) projects are *contextual* forward-citations, and the core argument survives even if their final numbers shift in review. We have confirmed this explicitly: no held or provisional number from those anchors enters the spine.

Accordingly, the manuscript is complete and reproduces green against its in-repository anchors, but **submission is held** behind an explicit release gate until the trust and value anchors publish (and ideally the action anchors as well). Reproduction and release are separate concerns: `reproduce.sh` regenerates every in-repository anchor; the release gate (`release_gate.py`, `submit.sh`) refuses to submit while either anchor is unpublished. The path to lift the hold, and the per-number re-check that fires when each anchor is accepted, are documented in `SUBMISSION_BLOCK.md` and `PROVISIONAL_LEDGER.md`.

Data and code availability

This is a synthesis; it generates no new data. The in-repository anchors are reproduced by `bash reproduce.sh`: a self-contained re-derivation of the CRLB identifiability wall (Fig. 1; `scripts/crlb_wall.py`), the D^* test–retest correlation interval (`scripts/retest_ci.py`, carrying the seeded BCa bootstrap CI and reproducing its Fisher-z twin), and the external D^*–K^{trans} evidence table (`scripts/dstar_ktrans.py`). The underlying test–retest data are the open ACRIN-6698 / I-SPY2 breast DWI cohort (TCIA, CC-BY-4.0 [3]). No private clinical data are touched. All manuscript numbers are auto-generated into `numbers.tex` by `paper/consistency.py` and trace to committed sources.

Forward-citation note. Fashion [1] and Minos [2] are cited as in-program forward-references and are unpublished at the time of writing; Gauge [5] and Lethe [4] are companion manuscripts. No published DOI is asserted for an unpublished anchor; each is swapped to its published reference at finalization per the checklist in `PROVISIONAL_LEDGER.md`.

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